Data Science

BY THE END:

You can explain what data science is and what data scientists do.

Overview

A mobile network sees that thousands of customers have stopped renewing their monthly bundles. The records show what happened, but the company needs to understand the pattern, test possible explanations, and decide what evidence would support a solution. This kind of open-ended problem is where **data science** becomes useful.

What is data science?

Data science combines statistics, programming, domain knowledge, and data analysis to extract insights and build data-driven solutions.

Data science may use statistical analysis, experiments, simulations, careful exploration, or machine learning.

Each part contributes something different:

PART	WHAT IT CONTRIBUTES
Statistics	Ways to measure patterns, uncertainty, and evidence
Programming	A way to work reliably with large or complex data
Domain knowledge	Understanding of the real-world problem and its constraints
Data analysis	Methods for exploring data and communicating what it shows

Knowing the business matters as much as knowing the numbers. A pattern in mobile data means little if the person studying it does not understand bundles, pricing, customer behaviour, or how the records were created.

What data scientists do

Data scientists often work on problems such as:

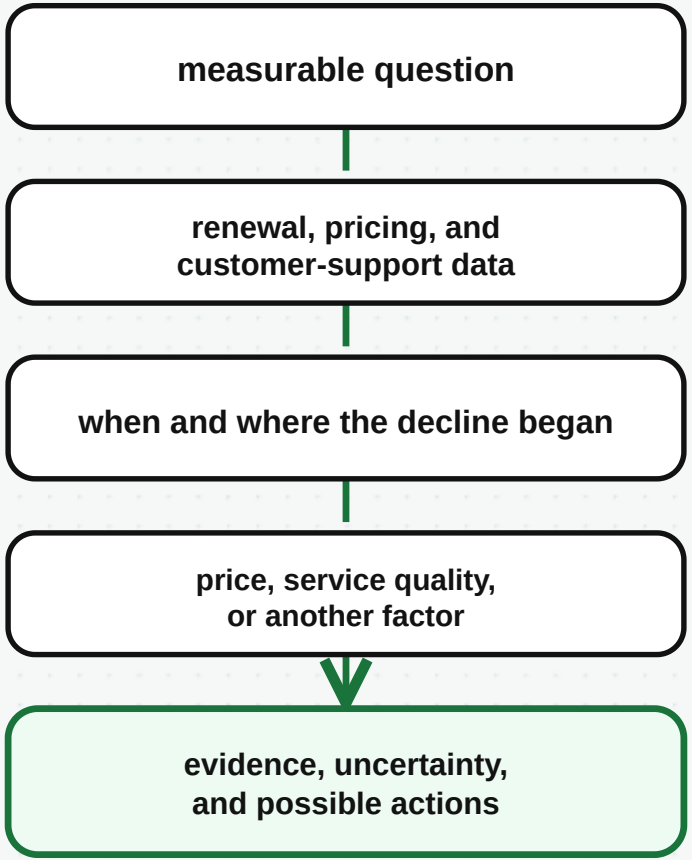
- Forecasting demand
- Detecting fraud
- Designing and evaluating experiments
- Investigating complex behaviour
- Building data products or predictive models

The work usually begins with a question, not a model. A data scientist decides what evidence is needed, checks whether suitable data exists, chooses an appropriate method, tests the result, and explains its limitations.

Example

For the falling renewals, a data scientist might:

1. Turn “customers are leaving” into a measurable question
2. Combine renewal, pricing, and customer-support data
3. Look for when and where the decline began
4. Test whether price, service quality, or another factor best explains it
5. Present the evidence, uncertainty, and possible actions



If the company later asks which individual customers are likely to leave next month, the problem has changed. It now needs a prediction. That is where machine learning may enter the project.